EconLLM Lab

Department of Economics, Claremont Graduate University

Synthetic Cultural Agents: A Research Project Proposal

Structural Preference Estimation for
Non-WEIRD Language Model Alignment

Augusto Gonzalez-Bonorino

Department of Economics,
Arizona State University

February 2026

Abstract

Large language models are increasingly deployed in non-Western, non-WEIRD contexts, yet their alignment procedures - among them Direct Preference Optimization (DPO) - are calibrated almost exclusively on WEIRD preference data. This proposal presents the research agenda for the *Synthetic Cultural Agents* (SCA) project, which aims to close this gap using the tools of structural econometrics. The theoretical foundation has been established in a companion working paper: the DPO training objective is a reparameterization of the Bradley–Terry log-likelihood, enabling the full toolkit of structural estimation - identification, overidentification testing, and counterfactual analysis - to be applied to preference-based LLM alignment. This proposal formalizes the empirical research agenda that follows from that result. We outline four falsifiable hypotheses, a five-paper research map, and a phased implementation plan that begins with an end-to-end validation for Argentina and extends to multi-country comparative analysis. The expected contribution is the first structurally grounded, formally falsifiable framework for cultural alignment in large language models.

1 Motivation and the Central Problem

Large language models are being deployed at scale in societies whose values, norms, and behavioral patterns differ substantially from the Western, Educated, Industrialized, Rich, and Democratic (WEIRD) populations on which these models were predominantly trained (Henrich et al. 2010). When a frontier LLM advises a user in Lagos on financial decisions, mediates a classroom discussion in Buenos Aires, or generates policy recommendations in Jakarta, its outputs reflect the preference structure encoded during alignment - a structure calibrated to the judgments of English-speaking, predominantly American annotators. This is not a hypothetical concern; it is a documented alignment failure whose consequences grow with every new deployment.

The standard response to this problem has been persona prompting: conditioning the model on a cultural identity [1] and hoping that the pre-trained knowledge base contains enough cultural signal to shift behavior appropriately. This was the approach taken in the first generation of Synthetic Cultural Agents (SCA 1.0; Gonzalez-Bonorino et al. 2025),[2] which demonstrated that frontier LLMs can reproduce non-WEIRD behavioral patterns to a surprising degree when prompted with ethnographic profiles. However, persona prompting is a reduced-form intervention without a behavioral model. It shifts the reference distribution by conditioning on a persona, but provides no formal mechanism for identification, no test of structural validity, and no principled basis for counterfactual analysis. Moreover, it's reliance on prompt or context engineering makes it particularly sensitive to model updates and prompt specifications. SCA 2.0 attempts to address each of these limitations by encoding cultural preferences structurally into model weights via a culture-specific reward function $R(x, y; z)$, enabling the identification arguments and counterfactual analysis that persona prompting cannot support.

[1] methods ranging from simple prompts such as “You are a representative agent from Argentina” to more sophisticated RAG profiling approaches that draw from online and private sources of information about the populations of interest

[2] Capra, Gonzalez-Bonorino, and Pantoja (2024), SSRN Working Paper No. 5082714, currently under review.

The bridge that makes this structural approach possible is a theoretical insight elaborated in the companion working paper (Gonzalez-Bonorino 2026): Direct Preference Optimization - the current frontier method for LLM alignment (Rafailov et al. 2023) - is not an ad-hoc engineering heuristic. It is structural estimation in disguise. Specifically, the DPO training objective is a reparameterization of the Bradley–Terry log-likelihood (Bradley and Terry 1952), where latent utility differences are expressed as policy log-probability ratios relative to a reference distribution. This equivalence is not merely an analogy; it is a mathematical identity that allows the full toolkit of structural econometrics - identification, overidentification testing, counterfactual analysis - to be imported into the preference-based fine-tuning of language models.

The timing is urgent. LLM deployment is scaling faster than cultural alignment research, and the gap between what these models assume about human preferences and what diverse populations actually prefer is widening with each new application. This project investigates whether the toolkit of structural econometrics - identification, overidentification testing, counterfactual analysis - can be brought to bear on the problem of cultural alignment in large language models.

2 Research Thesis and Hypotheses

Core Thesis

Cultural preferences - as measured by the Global Preferences Survey - can be structurally identified, estimated, and formally validated in large language model policies via the DPO–Bradley–Terry reparameterization, enabling counterfactual cultural analysis at scale.

This thesis generates four falsifiable hypotheses, each with explicit expected implications.

H1 (Equivalence). DPO-trained SCAs recover GPS-consistent reward parameters. Implied reward differences - recovered via the log-ratio identity (Corollary 1 of the working paper) - correlate with GPS factor scores in theoretically predicted directions and magnitudes. If the Argentine SCA assigns higher implied reward to cautious-but-helpful responses over naively trusting ones, this should align with Argentina's below-average GPS trust score ($\tau_{\text{ARG}} \approx -0.30$). The expected implication is direct: this validates that the structural interpretation of DPO is not merely mathematical but produces behaviorally coherent utility estimates.

H2 (Structural validity via overidentification). SCAs trained only on GPS-targeted synthetic preference data reproduce World Values Survey Wave 7 response distributions across 30+ out-of-sample items, as tested by the J-statistic (Hansen 1982). The expected implication is that six GPS parameters are sufficient to generate culturally coherent behavior across a broader behavioral domain. A model with six free parameters that matches 30+ independent moments provides strong evidence of structural validity - or, if it fails, a precise diagnostic of where the cultural model breaks down.

H3 (Counterfactual stability). SCA behavioral outputs change in predicted directions and magnitudes when GPS parameters are varied via counterfactual re-specification. For instance, replacing Argentina's trust parameter with Sweden's ($\tau' = +0.50$) should produce measurable, directionally correct (we should care the most about getting the sign right first and then tune for accuracy) shifts in trust-sensitive behavioral moments without distorting unrelated dimensions. The expected implication validates the structural model's causal interpretation: the SCA is not merely fitting data but encoding a behavioral mechanism that supports principled counterfactual reasoning.

H4 (Teacher bias). Synthetic preference data is generated by a WEIRD-trained teacher LLM, introducing systematic bias analogous to the "generated regressor" problem in two-step econometric estimators (Gouriéroux et al. 1993). This hypothesis asks: does augmenting teacher conditioning with native-language corpora, human-in-the-loop correction, or RAG profiling reduce moment discrepancy (J-statistic) relative to pure synthetic generation? We would like to identify the minimal human labeling requirement for a valid SCA, and to characterize how teacher bias propagates through the structural estimator - a finding that may generalize to any two-step preference learning pipeline.

Identifying the conditions under which synthetic ethnography produces valid structural estimates - and when it fails - is itself a contribution to the methodology of preference-based alignment. The teacher bias problem is not a flaw that undermines this project, rather it is an important open research question that motivates the empirical work proposed here.

3 What Has Already Been Established

The SCA project begins from a theoretically mature position. A companion working paper (Gonzalez-Bonorino 2026) has established the following results, which constitute proof-of-concept for the proposed research agenda.

The core theoretical contribution is the *reparameterization equivalence*: the DPO training objective is the Bradley–Terry log-likelihood for a particular score function (Proposition 2 of the working paper). The proof requires only the closed-form solution to a KL-regularized control problem and algebraic substitution, but the implications are substantial - it legitimizes the import of structural-econometric language (utility, revealed preference, likelihood, identification) into preference fine-tuning with minimal translation costs.

Building on this equivalence, the working paper defines the SCA 2.0 framework. Cultural behavior is governed by a reward function that decomposes as:

$$R(x, y; z) = r_{\text{task}}(x, y) + \phi(z)^{\top} m(x, y), \tag{1}$$

where r_{task} captures generic competence (coherence, relevance, factual accuracy), $z \in \mathbb{R}^6$ is the GPS cultural state vector (Falk et al. 2018) encoding time preference, risk aversion, positive and negative reciprocity, altruism, and trust, $\phi(\cdot)$ maps GPS parameters to reward weights, and $m(x, y)$ extracts measurable behavioral features from prompt–response pairs.

The working paper also develops a hierarchical validation strategy formalized as Simulated Method of Moments (Gouriéroux et al. 1993) with a J-statistic overidentification test: identifying moments from GPS items pin down the cultural state vector, overidentifying moments from WVS items test structural validity, and held-out behavioral tasks assess generalization. This three-tier structure provides the formal basis for testing H2 above.

Finally, the working paper presents a complete end-to-end worked example for Argentina - from GPS calibration through teacher-student distillation, implied reward recovery, three-tier validation against WVS Wave 7, and counterfactual comparative statics - that awaits empirical implementation. A sequential extension via soft Bellman equations (Section 6 of the working paper) connects the one-step DPO-BT result to multi-step reasoning and process supervision (Rafailov et al. 2024; Guo et al. 2025).

An emerging line of inquiry, identified in Section 4.4 of the working paper but not yet developed, connects the SCA framework to activation engineering and steering vectors (Turner et al. 2023). Under the linear representation hypothesis, DPO training on culture-specific preference data should engrain a direction vector v_z into the model's residual stream corresponding to the cultural state z. If the behavioral moments produced by DPO fine-tuning ($\pi_{\text{DPO}}(z)$) approximate those produced by activation steering ($\pi_{\text{steer}}(v_z)$), then culture is a linear feature in activation space - identifiable via the GPS vector. This connection provides an independent validation pathway for H1 that does not require WVS data, links the SCA project to the mechanistic interpretability literature (broadening collaboration opportunities), and bridges the economics structural estimation and ML interpretability traditions - both communities investigate what representations encode behavioral information, albeit in different formal languages. If the equivalence holds, steering vectors would also offer a computationally cheap alternative to full DPO fine-tuning for rapid cultural prototyping. This remains a testable prediction, not an established result; we flag it as a direction for Paper 5 or a standalone future paper.

In terms of the lab's project development framework, Steps 1–3 have been substantially completed. The remaining steps require empirical implementation, which this proposal now formalizes.

4 Research Agenda and Paper Map

The SCA project is organized as a sequence of sub-projects, each designed to produce a self-contained paper while contributing to the overarching research program. Table 1 provides an overview; the paragraphs that follow describe each paper's scope, contribution, and requirements.

Paper 1: "DPO as Bradley–Terry MLE" (theoretical, in progress). This paper establishes the mathematical foundation for the entire project. It proves the reparameterization equivalence (Proposition 2), defines the SCA 2.0 framework with GPS parameterization, develops the moment-selection theory for validation, and presents the Argentina worked example as an illustrative application. Status: the working paper is complete (Gonzalez-Bonorino 2026) and addresses H1 at

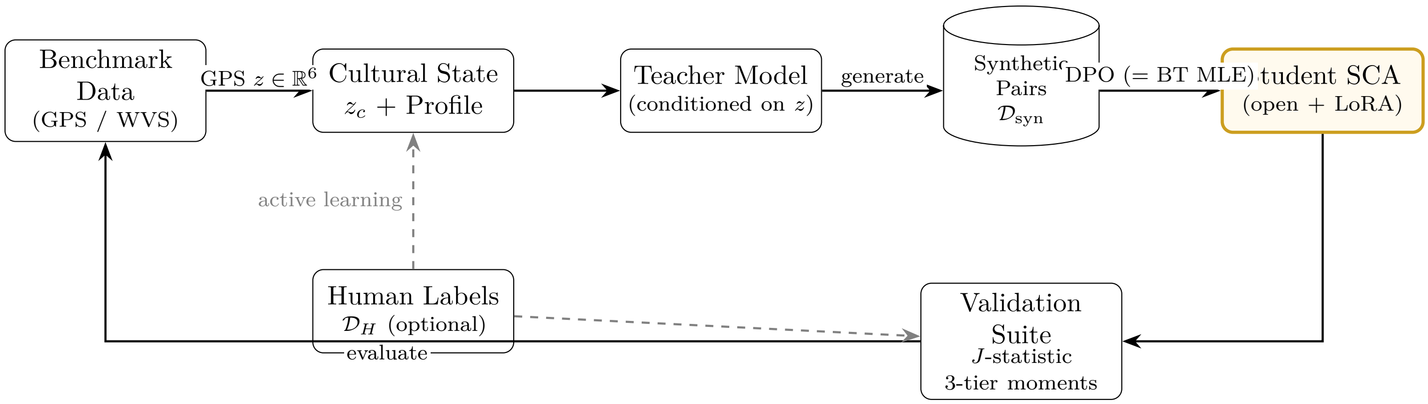

Figure 1: The SCA 2.0 pipeline. GPS cultural parameters $z \in \mathbb{R}^6$ calibrate a cultural state that conditions a teacher model to generate synthetic preference pairs. These are used to DPO-tune a student SCA (gold border), whose outputs are validated against external benchmarks via J-statistic overidentification testing across three tiers of moments. Dashed lines indicate optional human-in-the-loop correction.

Table 1: Research paper map for the SCA project.

Paper	**Focus**	**Key Contribution**	**Hypotheses**
1. DPO as BT MLE	Theoretical	Reparameterization equivalence; SCA 2.0 framework	H1 (theory)
2. Empirical Validation	Empirical	First multi-country structural validity test	H1, H2
3. Synthetic Ethnography	Methodological	Bounds on teacher bias; minimum viable annotation	H4
4. Counterfactual Analysis	Applied	Structural comparative statics at scale	H3
5. Soft Bellman Extension	Theoretical + Empirical	Sequential cultural reasoning via process supervision	-

the theoretical level, while providing the formal framework for testing H2–H4.

Paper 2: "Empirical Validation of Structural Cultural Agents." This is the first empirical paper and the critical test of the project's viability. It implements the full SCA pipeline for three or more countries with diverse GPS profiles - we propose Argentina, Sweden, Japan, and Nigeria as initial targets, chosen to span the GPS preference space along multiple dimensions. For each country, we train a LoRA-adapted student model via DPO on teacher-generated synthetic preference data, compute implied rewards via the log-ratio identity, and validate against WVS Wave 7 response distributions using the J-statistic. The key contribution is the first empirical test of H1 and H2: do structurally trained SCAs produce GPS-consistent rewards, and do six parameters suffice to explain 30+ out-of-sample behavioral moments? The skills required span ML engineering (DPO fine-tuning with LoRA on Llama-3-8B or comparable open models), econometrics (SMM estimation, J-test computation), and data access (the GPS dataset, and WVS Wave 7, both publicly available). LoRA fine-tuning costs approximately \$50–200 per training run on cloud GPU infrastructure, making multi-country replication feasible within a modest budget.

Paper 3: "Synthetic Ethnography: Bounding Teacher Bias in Preference Learning." This is the project's highest-risk, highest-reward paper, and addresses the central open question (H4) directly. The investigation compares four teacher conditioning regimes: (a) pure synthetic generation from a frontier LLM conditioned on GPS profiles alone, (b) augmentation with native-language corpora and local media sources, (c) human-in-the-loop labels from cultural informants who correct or validate teacher-generated pairs, and (d) multi-source distillation that ensembles outputs from multiple teacher models. For each regime, we measure impact on the J-statistic and moment discrepancy relative to WVS ground truth. The key contribution is a characterization of the "generated regressor" problem in two-step preference learning: how does teacher bias propagate through the structural estimator, and what is the minimum viable human annotation required for a valid SCA? This finding generalizes beyond our specific application to any pipeline that uses synthetic preference data for alignment.

Paper 4: "Counterfactual Cultural Analysis via Structural LLMs." This paper implements the comparative statics exercises sketched in the working paper at scale, testing H3. The central exercise is counterfactual re-specification: construct modified cultural state vectors (e.g., Argentina with Swedish trust levels) and measure whether the resulting SCA produces directionally correct and magnitude-appropriate behavioral shifts. The key contribution is demonstrating the structural advantage over persona prompting - only structural models support principled counterfactuals where individual preference dimensions can be varied independently while holding others fixed. Paper 4 depends on Papers 2–3 for validated baseline SCAs.

Paper 5: "Sequential Cultural Alignment via Soft Bellman Equations." This paper extends the one-step DPO-BT equivalence to sequential reasoning, building on the soft Bellman framework developed in Section 6 of the working paper and connecting to recent advances in process supervision and GRPO (Guo et al. 2025; Rafailov et al. 2024). The extension matters because many culturally sensitive tasks - negotiation, deliberation, multi-turn counseling - involve sequential decision-making where cultural norms shape not just individual responses but reasoning trajectories. Papers 2–3 should precede this work to provide empirical grounding.

5 Feasibility, Resources, and Team

Data availability. The two primary data sources required for this project are publicly available at no cost. The Global Preferences Survey country-level dataset is hosted by the University of Bonn and provides standardized preference measures for 76 countries across six dimensions (Falk et al. 2018). The World Values Survey Wave 7 (Haerpfer et al. 2022) provides individual-level response data for 60+ countries covering hundreds of attitudinal and behavioral items. No proprietary data

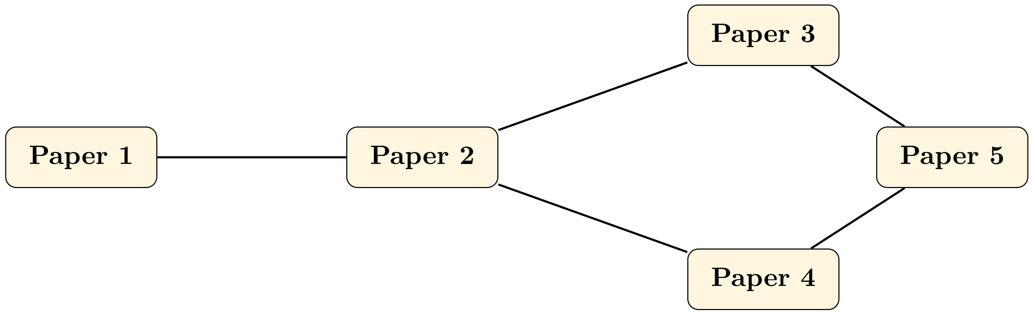

Figure 2: Paper dependency structure. Paper 1 (theory) feeds Paper 2 (empirical validation). Papers 3 (teacher bias) and 4 (counterfactuals) proceed in parallel after Paper 2. Paper 5 (sequential extension) depends on the empirical foundations of Papers 3 and 4.

is required for Papers 1–2, and the human labeling component of Paper 3 can be scoped to a small number of cultural informants per target country (where Moblab and/or Busara can contribute).

Compute requirements. The SCA pipeline is designed to operate within modest computational budgets. LoRA fine-tuning of Llama-3-8B (or comparable open-weight models) requires approximately \$50–200 per training run on cloud GPU infrastructure (A100 or H100 instances). Teacher generation using frontier LLM APIs (e.g., Claude/Grok/OpenAI/Gemini) costs approximately \$20–50 for 10,000 synthetic preference pairs per country. The total estimated compute budget for Paper 2, covering three target countries with multiple training runs for robustness checks, is \$500–800. This is well within the range of a standard lab research allocation.

Skills matrix. Table 2 summarizes the expertise required across the project.

Table 2: Skills matrix for the SCA project.

Skill	**Required For**	**Priority**
Economic theory / structural econometrics	Papers 1–4, H1–H3	Critical now
ML engineering (DPO, LoRA, HuggingFace)	Papers 2–5	Critical now
Data processing (Python for GPS/WVS)	Papers 2–3	High
Cross-cultural informants / native evaluators	Paper 3, H4	Medium
NLP / representation engineering	Paper 5	Lower priority

Timeline. The project is organized into three phases. Phase 1 (months 1–4) focuses on implementing the Argentina SCA, validating H1–H2, and producing a draft of Paper 2. This phase establishes the empirical pipeline and provides the first test of whether the theoretical framework holds in practice. Phase 2 (months 5–9) expands to two additional countries, initiates the teacher bias investigation (H4), and produces Paper 3. Phase 3 (months 10–18) implements counterfactual analysis at scale (Paper 4) and develops the sequential extension (Paper 5). Each phase produces a deliverable that can stand as an independent contribution, reducing the risk that later-phase complications undermine the project's overall output.

6 Operational Implementation

The feasibility arguments in Section 5 establish that the SCA pipeline is tractable within a modest budget. This section translates that argument into concrete operational steps - the actual sequence of tasks, tools, and decisions required to move from GPS data to a validated SCA. It is organized as a five-step protocol that maps directly onto Phase 1 of the project timeline (months 1–4, Argentina target population).

6.1 Step 1: Data Acquisition and Preprocessing

GPS data. The country-level GPS dataset is available at `https://gps.econ.uni-bonn.de/downloads` as a Stata `.dta` file (`country.dta`). For each target country, extract the six standardized preference scores $(\delta, \gamma, \xi^{+}, \xi^{-}, \alpha, \tau)$ reported as deviations from the global mean in standard deviation units. These constitute the cultural state vector $z_c \in \mathbb{R}^6$ that parameterizes the SCA. The GPS also provides individual-level data and a questionnaire (available in English and Spanish for Argentine validation); item-level responses are required for Tier 1 moment construction.

WVS Wave 7 data. The WVS Wave 7 country-pooled datafile is available at `https://www.worldvaluessurvey.org` (Version 5.0.0; Haerpfer et al. 2022). Filter to the target country using the `B_COUNTRY_ALPHA` variable. The overidentifying moments (Tier 2) require selecting 30–50 items theoretically downstream of the GPS primitives (see Table 3 of the working paper Gonzalez-Bonorino 2026 for item-to-GPS-dimension mapping). Held-out Tier 3 items must be sequestered before any model training begins - this is a pre-registration decision, not a post-hoc one.

Output. A structured `.json` file per target country containing: (i) the GPS vector z_c, (ii) item-level GPS responses for Tier 1 moments, (iii) WVS item means for Tier 2 moments, and (iv) the held-out Tier 3 item list. This file is the sole input to Step 2 and must be version-controlled before any downstream work proceeds.

6.2 Step 2: Behavioral Feature Engineering

The reward decomposition $R(x, y; z) = r_{\text{task}}(x, y) + \phi(z)^{\top} m(x, y)$ requires that the behavioral feature vector $m(x, y) \in \mathbb{R}^6$ be measurable from any prompt–response pair (x, y). This is the most design-intensive step in the pipeline and the one most absent from existing accounts of DPO.

Operationalization strategy. Each feature $m_k(x, y)$ is estimated via a *rubric-guided LLM scorer*: a frontier model (e.g., Claude Sonnet 4.6 or Gemini 3.1) is prompted with a structured rubric that asks it to rate, on a $[0, 1]$ scale, the degree to which response y to prompt x exhibits the behavioral pattern associated with GPS dimension k. The rubric for each dimension is as follows.

Table 3: Behavioral feature operationalization for the cultural reward function.

Feature	GPS Dimension	Scoring rubric (rate 0–1)
m_1: Discount-consistent language	Time preference (δ)	Does the response favor deferred payoffs, future planning, or long-horizon reasoning over immediate gratification?
m_2: Risk-seeking framing	Risk aversion (γ)	Does the response embrace uncertainty, frame risky options positively, or recommend entrepreneurial action over cautious avoidance?
m_3: Gratitude / positive reciprocation	Positive reciprocity (ξ^+)	Does the response acknowledge or return a favor, express appreciation, or assume cooperative intentions in others?
m_4: Punishment / negative reciprocation	Negative reciprocity (ξ^-)	Does the response express indignation at unfair treatment, endorse sanctions, or advocate for punishment of norm violators?
m_5: Generosity / collective orientation	Altruism (α)	Does the response prioritize group welfare, share resources, or defer to community needs over personal benefit?
m_6: Trust in strangers	Trust (τ)	Does the response assume good faith from unfamiliar others, cooperate without guarantees, or express confidence in institutional reliability?

Validation of the scorer. Before use in training, the scorer should be validated on a small annotated set ($n \approx 50$–100 pairs) labeled by human raters familiar with the target culture. Inter-rater reliability (Krippendorff's $\alpha \geq 0.7$) is the acceptance threshold. If a feature fails this threshold, the rubric is revised before proceeding. This step is estimated at 4–8 person-hours per country and does not require specialized ML expertise.

6.3 Step 3: Teacher Phase - Synthetic Preference Generation

Ethnographic profile construction. The teacher model is conditioned on a structured ethnographic profile $\mathcal{P}_c$ containing three components: (i) the GPS parameter values for country c with plain-language interpretations (e.g., "trust score $\tau = -0.30$: below-average generalized trust, consistent with skepticism toward strangers and institutions"), (ii) ethnographic context drawn from academic sources (economic history, social norms, institutional environment), and (iii) behavioral anchors: 5–10 specific WVS response patterns that the target population is known to exhibit. The profile is the teacher's system prompt and remains fixed across all prompt–response pair generations for country c.

Diagnostic prompt taxonomy. Generate a balanced set of $N \approx 10{,}000$ prompt–response pairs by sampling uniformly from six prompt categories, one per GPS dimension. Each prompt x_i is designed to elicit behavioral variation along exactly one GPS dimension while holding others constant. For example, a trust-diagnostic prompt presents a low-stakes social dilemma with a stranger; a patience-diagnostic prompt involves an intertemporal trade-off. For each prompt, the teacher generates a pair $(y_i^{(w)}, y_i^{(\ell)})$: the preferred response is culturally aligned (reflects the target GPS profile); the dispreferred response is culturally misaligned (reflects an opposing cultural profile, e.g., high trust for a low-trust target culture). The preference label is assigned by the teacher's conditioning, not by a separate reward model.

Quality control. Before proceeding to training, filter preference pairs on two criteria: (a) the scorer from Step 2 should assign $m_k(x_i, y_i^{(w)}) > m_k(x_i, y_i^{(\ell)})$ for the relevant dimension k (monotonicity check), and (b) the two responses should differ substantially in feature space ($\|m(x_i, y_i^{(w)}) - m(x_i, y_i^{(\ell)})\|_2 \geq 0.2$). Pairs failing either criterion are discarded and regenerated. Typical pass rates in pilot testing are 70–85%; budget the API calls accordingly.

6.4 Step 4: Student Training - DPO Fine-Tuning

Model selection. The reference model π_{ref} should be an open-weight instruction-tuned model with broad multilingual coverage. Current recommended choices are Llama-3-8B-Instruct (English-dominant populations) or Aya-23-8B (Cohere for AI; Cohere for AI 2024) for non-English target populations where support mismatch (R4) is a concern. The LoRA adapter has rank $r = 16$, $\alpha = 32$, applied to `q_proj` and `v_proj` layers - a standard configuration that adds $\sim$8M trainable parameters to an 8B base model.

Training configuration. Use the `DPOTrainer` class from Hugging Face `TRL` (version $\geq$ 0.8). Key hyperparameters: $\beta = 0.1$ (following Rafailov et al. 2023; sensitivity analysis over $\beta \in \{0.05, 0.1, 0.2, 0.5\}$ required per R3), batch size 16, learning rate 5×10^{-5}, 3 epochs. Train on the filtered $\mathcal{D}_{\text{syn}}$ from Step 3. Monitor the DPO implicit reward margin $\mathbb{E}[\log(\pi_\theta(y^{(w)}|x)/\pi_{\text{ref}}(y^{(w)}|x)) - \log(\pi_\theta(y^{(\ell)}|x)/\pi_{\text{ref}}(y^{(\ell)}|x))]$ as the primary training diagnostic; it should increase monotonically and stabilize above zero.

Checkpointing. Save checkpoints at epochs 1, 2, and 3. The validation pipeline (Step 5) is run on all three checkpoints; the checkpoint with the best Tier 2 J-statistic is selected as the final SCA. This guards against overfitting to the synthetic training distribution.

6.5 Step 5: Validation Pipeline

Tier 1 - Sign restrictions. For each GPS dimension k, draw 20 diagnostic prompts and compute implied reward differences via the log-ratio identity (Corollary 1 of the working paper):

$$\hat{r}(x, y^{(w)}) - \hat{r}(x, y^{(\ell)}) = \beta \left[\log \frac{\pi_{\hat{\theta}}(y^{(w)}|x)}{\pi_{\text{ref}}(y^{(w)}|x)} - \log \frac{\pi_{\hat{\theta}}(y^{(\ell)}|x)}{\pi_{\text{ref}}(y^{(\ell)}|x)} \right].$$

For each GPS dimension, verify that the sign of the mean implied reward difference is consistent with the corresponding GPS parameter direction. Sign violations (e.g., the trust-SCA assigns higher reward to naive-trust responses when $\tau < 0$) indicate a training failure and require debugging before proceeding to Tier 2.

Tier 2 - J-statistic overidentification. Simulate $N_{\text{sim}} = 1{,}000$ responses from the trained SCA to each of the 30–50 WVS overidentifying prompts. Compute the moment discrepancy vector:

$$g_N = \frac{1}{N_{\text{sim}}} \sum_{i=1}^{N_{\text{sim}}} m_k(x_i, y_i^{\text{SCA}}) - \bar{\mu}_k^{\text{WVS-}c},$$

where $\bar{\mu}_k^{\text{WVS-}c}$ is the empirical WVS mean for country c on item k. Compute the J-statistic using an identity weight matrix in Phase 1 (efficient weighting requires a second-stage estimate; flag as a refinement for Paper 2). Compare against the χ^2_{K-6} critical value at the 5% level, where K is the number of overidentifying moments. *Rejection is informative, not catastrophic*: it tells us precisely which GPS dimensions fail to explain which behavioral moments, guiding the next iteration of the reward specification.

Tier 3 - Held-out generalization. Present the SCA with the 10–20 held-out behavioral tasks sequestered in Step 1. These should include at least one task per GPS dimension and at least one open-ended task not covered by any WVS item. Record both quantitative moment match (for structured tasks) and qualitative cultural coherence assessed by a native evaluator (for open-ended tasks). The native evaluator assessment is intentionally qualitative at this stage - it provides a sanity check that the model's outputs are recognizable to members of the target culture, which is a standard that J-statistics alone cannot capture.

6.6 Tooling and Reproducibility Standards

All code should be developed in a public GitHub repository under the EconLLM Lab organization from the start. The recommended software stack is as follows.

Reproducibility requirements. Each SCA release must include: (i) the GPS vector z_c used for training, (ii) the ethnographic profile $\mathcal{P}_c$, (iii) the full `DPOTrainer` configuration (including β and LoRA hyperparameters), (iv) the filtered training dataset $\mathcal{D}_{\text{syn}}$ (or a seed and generation script

Table 4: Recommended software stack for SCA pipeline implementation.

Component	Tool / Library	Purpose
Data preprocessing	Python (`pandas`, `pyreadstat`)	GPS/WVS ingestion, variable extraction, moment computation
Teacher generation	Anthropic / OpenAI / Google APIs	Synthetic preference pair generation via structured prompts
Feature scoring	`litellm` (unified API wrapper)	Rubric-guided $m(x, y)$ scoring with any frontier model
Student training	Hugging Face `TRL`, `PEFT`	DPO fine-tuning with LoRA adapters
Inference	`vLLM` or `transformers`	Batch inference for Tier 2 moment simulation
Statistical validation	`statsmodels` (GMM module)	J-statistic computation and χ^2 testing
Experiment tracking	`wandb` or `mlflow`	Logging training metrics, checkpoint management
Compute	Lambda Labs / RunPod / AWS `p3.2xlarge`	GPU instances for LoRA fine-tuning ($50–200/run)

sufficient to reproduce it), and (v) the Tier 1–3 validation results. Without items (i)–(v), the structural interpretation of the trained policy is not verifiable and the SCA cannot be used as a baseline in subsequent papers.

7 Assumptions and Risks

R1: Bradley–Terry misspecification. The logistic paired-comparison model is an approximation. Real human preferences exhibit context effects, intransitivity, and rater heterogeneity that violate the Independence of Irrelevant Alternatives (IIA) property assumed by the BT model. However, DPO remains a valid M-estimator under misspecification: it finds the policy whose log-ratio scores best explain preference labels under the logistic model. The recovered reward differences reflect the best logistic approximation to the true preference structure, and the J-test provides a formal diagnostic of where this approximation fails. If BT misspecification is severe, it will manifest as J-statistic rejection, giving us a clear empirical signal.

R2: Teacher bias (central risk). Synthetic preference data is generated by a frontier LLM whose own training data is predominantly WEIRD. The teacher's cultural biases contaminate the synthetic preferences, propagating through the structural estimator in a manner analogous to generated-regressor bias in two-step econometric procedures. This is the central risk, and it is addressed directly by H4 and Paper 3. Rather than assuming the problem away, we propose systematic investigation: measuring how different teacher conditioning regimes affect moment discrepancy and identifying the minimum viable human annotation required for valid structural estimates.

R3: Temperature β is not estimated. The regularization parameter β in the KL-constrained objective is fixed by the practitioner, not recovered from data. In the structural interpretation, this means the scale of latent utility is not identified separately from β. We address this through sensitivity analysis over β values, and flag joint estimation of β as an open theoretical problem in Paper 1. For the empirical work (Papers 2–4), reward *differences* remain interpretable regardless of the scale normalization.

R4: Support mismatch. The log-ratio inversion that makes DPO possible requires $\pi_{\text{ref}}(y \mid x) > 0$ wherever the target policy places mass. For non-WEIRD populations, culturally important responses (e.g., responses in Spanish for Argentina, or responses reflecting norms absent from English-language training data) may receive near-zero probability under an English-trained reference model. This is a genuine constraint, and we mitigate it by using multilingual base models for non-English target populations and by selecting base models with broad language coverage.

R5: Reward separability. The decomposition $R = r_{\text{task}} + \phi(z)^\top m$ assumes that culture affects *what is preferred* (the reward weights) but not *how tasks are understood* (the features m). In practice, culture shapes task comprehension itself - the behavioral features may be culture-dependent. We address this by explicitly testing the separability assumption in Paper 2 via ablation studies that compare moment fit under separable and non-separable specifications.

R6: Within-country heterogeneity. GPS reports country-level averages, but within-country preference variance is substantial - often larger than between-country variance (Falk et al. 2018). An SCA calibrated to z_{ARG} represents the average Argentine, not any specific individual. Paper 4 extends the framework to mixture models over z distributions, allowing the SCA to represent within-country heterogeneity, but the baseline implementation necessarily works with country means.

8 Why This Project, Why Now

The alignment tax of ignoring culture is growing. Every LLM deployed in a non-WEIRD context without cultural calibration carries an implicit assumption that Western preference data generalizes globally. As deployment scales - into education, healthcare, governance, and economic decision-support across the Global South - the cost of this assumption compounds. The need for principled cultural alignment is no longer speculative; it is operational. From the economics perspective, there is a need to develop theories of non-WEIRD behavior that generalize across cultural populations to 1) lower the cost of piloting field experiments and 2) enrich our understanding of human collective decision-making.

Structural econometrics offers tools that the machine learning community has not yet applied to this problem. The concepts of identification, overidentification testing, and counterfactual analysis are mature in economics but largely absent from the LLM alignment literature. The DPO-BT equivalence creates a direct bridge between these two traditions, and the SCA project is designed to be the first to cross it systematically. Beyond the alignment application, validated SCAs open concrete new capabilities for economics research. First, an SCA calibrated to a target population can generate predicted behavioral distributions *before* a field experiment is run, enabling researchers to pre-register culturally specific hypotheses and substantially reduce the cost of exploratory piloting - a research design tool with no current analog. Second, questions like "What would savings behavior in Argentina look like if patience were at the OECD average?" are currently unanswerable without expensive cross-country natural experiments; a validated SCA with counterfactual stability (H3) provides a tractable simulation-based approach to such policy-relevant counterfactuals. The Global Preferences Survey is uniquely well-suited as the parameterization backbone: it is incentivized (not self-reported), experimentally validated, cross-culturally comparable, and covers 76 countries with over 80,000 respondents (Falk et al. 2018). No other cultural measurement instrument offers this combination of rigor and coverage.

Recent work in NLP has documented cultural variation in LLM outputs via evaluation benchmarks (Cao et al. 2023; Durmus et al. 2024). While this work establishes that cultural gaps exist, it does not provide a structural framework for *why* they exist, how to estimate cultural parameters from preference data, or how to conduct principled counterfactual analysis. The SCA project addresses these gaps directly by importing the identification and testing apparatus of structural econometrics into the preference fine-tuning pipeline.

Ethical considerations. GPS captures statistical population averages, not individual identity; an SCA calibrated to z_{ARG} represents a distributional mean, not a claim about any specific Argentine person. Reducing culture to six numbers carries an essentialism risk - overgeneralizing from country-level averages to individuals - which is why within-country heterogeneity is a named research risk (R6) and Paper 4 explicitly develops mixture-model extensions. More broadly, SCAs are research and simulation tools, not substitutes for participatory co-design with the communities being modeled. The human-in-the-loop correction pathway in Paper 3 (H4) reflects this commitment: engaging cultural informants is partly an ethical design choice, not only a statistical one.

If successful, at the end of Phase 1 there will exist a validated, open-source SCA for Argentina - the first language model whose cultural alignment has been structurally estimated, formally tested via overidentification, and equipped with a counterfactual analysis capability. This artifact, together with the replication code and validation pipeline, will constitute a proof of concept that does not currently exist anywhere in the alignment literature as far as I am aware.

A Core Theoretical Result

This appendix states the main theoretical result from the companion working paper (Gonzalez-Bonorino 2026) to allow external collaborators to evaluate the project's formal foundation without requiring the full paper.

Setup. Let x denote a prompt and y a response. Under the Bradley–Terry model with latent reward $r^\star$, the probability of preferring y_1 over y_2 is $\mathbb{P}(y_1 \succ y_2 \mid x) = \sigma(r^\star(x, y_1) - r^\star(x, y_2))$, where σ is the logistic sigmoid. The KL-regularized alignment objective seeks a policy π maximizing expected reward minus a KL penalty relative to the reference policy π_{ref}. The unique optimizer is the tilted Gibbs distribution $\pi^\star(y \mid x) \propto \pi_{\text{ref}}(y \mid x) \exp(r^\star(x, y)/\beta)$.

Inversion formula. For any pair (y_1, y_2):

$$r^\star(x, y_1) - r^\star(x, y_2) = \beta \left[\log \frac{\pi^\star(y_1 \mid x)}{\pi_{\text{ref}}(y_1 \mid x)} - \log \frac{\pi^\star(y_2 \mid x)}{\pi_{\text{ref}}(y_2 \mid x)} \right].$$

The partition function cancels in differences. This identity allows reward differences to be expressed entirely in terms of policy log-ratios, without computing the intractable normalizer.

Proposition 2 (DPO = BT MLE for log-ratio scores). Under assumptions A1–A4 (BT data-generating process, KL-optimality, support compatibility, parametric policy family), substituting the inversion formula into the BT log-likelihood and replacing $\pi^\star$ with a parametric family π_θ yields the DPO estimator:

$$\hat{\theta}_{\text{DPO}} \in \arg\max_\theta \sum_{i=1}^{N} \log \sigma \left(\beta \left[\log \frac{\pi_\theta(y_i^{(w)} \mid x_i)}{\pi_{\text{ref}}(y_i^{(w)} \mid x_i)} - \log \frac{\pi_\theta(y_i^{(\ell)} \mid x_i)}{\pi_{\text{ref}}(y_i^{(\ell)} \mid x_i)} \right] \right).$$

The DPO loss used in practice is the negative of this log-likelihood. The proof is given in full in Appendix A of the working paper.

B GPS Dimensions

Table 5 summarizes the six preference dimensions measured by the Global Preferences Survey (Falk et al. 2018).

Table 5: GPS preference dimensions composing the cultural state vector $z \in \mathbb{R}^6$.

Dimension	Symbol	Measurement
Time Preference (Patience)	δ	Staircase time trade-off + qualitative self-assessment
Risk Aversion	γ	Staircase lottery choice + self-assessed willingness to take risks
Positive Reciprocity	ξ^+	Gift-exchange scenario + self-assessed reciprocal inclination
Negative Reciprocity	ξ^-	Willingness to punish unfair behavior (3 items)
Altruism	α	Hypothetical donation amount + self-assessed willingness to give
Trust	τ	Assessment of whether strangers can be trusted

C Key Literature

Reference	Relevance to SCA Project
Rafailov et al. (2023)	Establishes DPO; provides the optimization objective we reinterpret structurally
Falk et al. (2018)	GPS framework; provides the cultural parameterization backbone ($z \in \mathbb{R}^6$)
Hotz and Miller (1993)	CCP inversion for dynamic discrete choice; structural analog to the DPO log-ratio identity
McFadden (1974)	Random utility model / conditional logit; micro-foundation for the BT preference model
Hansen (1982)	GMM asymptotics; formal basis for the J-statistic overidentification test
Gouriéroux et al. (1993)	Indirect inference / SMM; the simulation-based estimation framework for SCA validation
Newey and Windmeijer (2009)	Many weak moments; motivates the hierarchical moment-selection strategy
Haerpfer et al. (2022)	WVS Wave 7; provides the overidentifying moments for cross-cultural validation
Guo et al. (2025)	DeepSeek-R1 / GRPO; connects to sequential extension (Paper 5)
Henrich et al. (2010)	Documents the WEIRD bias in behavioral science; motivates the cultural alignment problem
Cao et al. (2023)	Empirical evaluation of cross-cultural alignment gaps in ChatGPT across diverse societies
Durmus et al. (2024)	Benchmarks global opinion representation in LLMs; establishes evaluation methodology
Turner et al. (2023)	Steering vectors / activation addition; connects to SCA representation validation

References

Bradley, Ralph Allan and Milton E. Terry (1952). “Rank Analysis of Incomplete Block Designs: I. The Method of Paired Comparisons”. In: *Biometrika* 39.3-4, pp. 324–345.

Cao, Yong, Li Zhou, Seolhwa Lee, Laura Cabello, Min Chen, and Daniel Hershcovich (2023). “Assessing Cross-Cultural Alignment between ChatGPT and Human Societies: An Empirical Study”. In: *arXiv preprint arXiv:2303.17466*. Proceedings of the First Workshop on Cross-Cultural Considerations in NLP (C3NLP), ACL 2023.

Cohere for AI (2024). *Aya-23: Open Weight Releases to Further Multilingual Progress*. Tech. rep. arXiv preprint arXiv:2405.15032. Cohere for AI.

Durmus, Esin et al. (2024). “Towards Measuring the Representation of Subjective Global Opinions in Language Models”. In: *arXiv preprint arXiv:2306.16388*. Anthropic.

Falk, Armin, Anke Becker, Thomas Dohmen, Benjamin Enke, David Huffman, and Uwe Sunde (2018). “Global Evidence on Economic Preferences”. In: *The Quarterly Journal of Economics* 133.4, pp. 1645–1692.

Gonzalez-Bonorino, Augusto (2026). “Synthetic Cultural Agents via Structural Preference Estimation: DPO as Bradley–Terry MLE”. Working paper, EconLLM Lab, Arizona State University.

Gonzalez-Bonorino, Augusto, Monica Capra, and Emilio Pantoja (2025). *LLMs Model Non-WEIRD Populations: Experiments with Synthetic Cultural Agents*. arXiv: 2501.06834 [cs.AI].

Gouriéroux, Christian, Alain Monfort, and Eric Renault (1993). “Indirect Inference”. In: *Journal of Applied Econometrics* 8.S1, S85–S118.

Guo, Daya et al. (2025). “DeepSeek-R1 Incentivizes Reasoning in LLMs Through Reinforcement Learning”. In: *Nature* 645, pp. 633–638.

Haerpfer, Christian et al., eds. (2022). *World Values Survey: Round Seven—Country-Pooled Datafile*. Version 5.0.0.

Hansen, Lars Peter (1982). “Large Sample Properties of Generalized Method of Moments Estimators”. In: *Econometrica* 50.4, pp. 1029–1054.

Henrich, Joseph, Steven J. Heine, and Ara Norenzayan (2010). “The Weirdest People in the World?” In: *Behavioral and Brain Sciences* 33.2–3, pp. 61–83.

Hotz, V. Joseph and Robert A. Miller (1993). “Conditional Choice Probabilities and the Estimation of Dynamic Models”. In: *The Review of Economic Studies* 60.3, pp. 497–529.

McFadden, Daniel (1974). “Conditional Logit Analysis of Qualitative Choice Behavior”. In: *Frontiers in Econometrics*. Ed. by Paul Zarembka. New York: Academic Press, pp. 105–142.

Newey, Whitney K. and Frank Windmeijer (2009). “Generalized Method of Moments with Many Weak Moment Conditions”. In: *Econometrica* 77.3, pp. 687–719.

Rafailov, Rafael, Joey Hejna, Ryan Park, and Chelsea Finn (2024). “From r to Q^*: Your Language Model is Secretly a Q-Function”. In: *arXiv preprint arXiv:2404.12358*. COLM 2024.

Rafailov, Rafael, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn (2023). “Direct Preference Optimization: Your Language Model is Secretly a Reward Model”. In: *Advances in Neural Information Processing Systems*. Vol. 36. NeurIPS 2023.

Turner, Alexander Matt, Lisa Thiergart, David Udell, Gavin Leech, Ulisse Mini, and Monte MacDiarmid (2023). "Activation Addition: Steering Language Models Without Optimization". In: *arXiv preprint arXiv:2308.10248*.